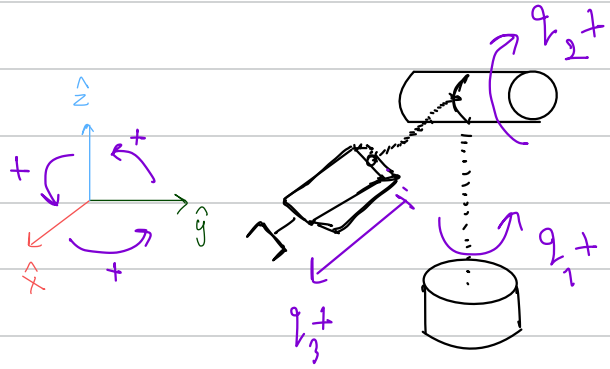
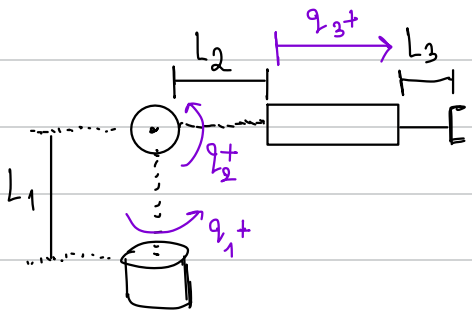
Kinematic Lab My Robot - 29 oct.

Isometric View



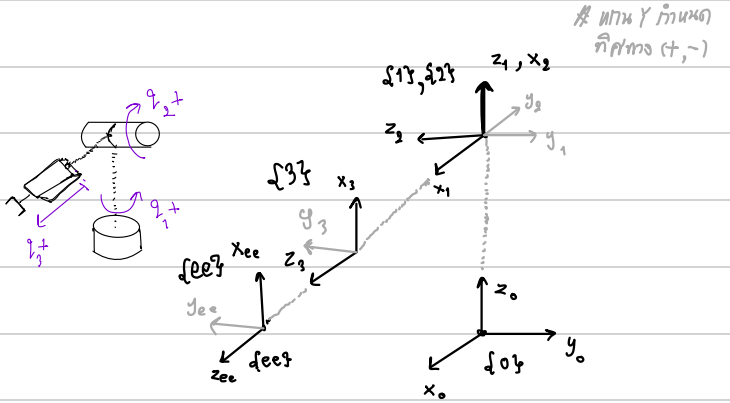
XZ - Plane



<Home Configuration>

Frame Assignment

based on Isometric view.



MDH Table.

i	i - 1	a_{i-1}	α_{i-1}	d_i	θ_i
0	1	0	0	L_1	q_1
1	2	0	$\frac{\pi}{2}$	0	$q_2 + \frac{\pi}{2}$
2	3	0	$\frac{\pi}{2}$	L_2+q_3	0
3	ee	0	0	L_3	0

Forward Kinematic Equation.

$T_{ee}^0 = T_1^0 T_2^1 T_3^2 T_{ee}^3$ $\quad C(\frac{\pi}{2}) = 0 \quad S(\frac{\pi}{2}) = 1 \quad C(\frac{\pi}{2}+q) = \quad S(\frac{\pi}{2}+q) =$

$$= \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & (-L_2+q_3) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -c_1 s_2 & s_1 & c_1 c_2 & (L_2+L_3+q_3)c_1 c_2 \\ -s_1 s_2 & -c_1 & s_1 c_2 & (L_2+L_3+q_3)s_1 c_2 \\ c_2 & 0 & s_2 & L_1+(L_2+L_3+q_3)s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

FK ⟶ IK

29 Oct

Pose Orientation.

$$\begin{bmatrix} -c_1 s_2 & s_1 & c_1 c_2 \\ -s_1 s_2 & -c_1 & s_1 c_2 \\ c_2 & 0 & s_2 \end{bmatrix}$$

Pose Position

$$X = (L_2+L_3+q_3)c_1c_2$$
$$y = (L_2+L_3+q_3)s_1c_2$$
$$z = L_1+(L_2+L_3+q_3)s_2$$

IK: let $R = L_2+L_3+q_3$

$$X = (L_2+L_3+q_3)c_1c_2 \quad \blacktriangleright \quad X = Rc_1c_2 \quad -①$$
$$y = (L_2+L_3+q_3)s_1c_2 \quad \blacktriangleright \quad Y = Rs_1c_2 \quad -②$$
$$z = L_1+(L_2+L_3+q_3)s_2 \quad \blacktriangleright \quad z = L_1+Rs_2 \quad -③$$

$$\blacktriangleright \quad z-L_1 = Rs_2 \quad -④$$

จาก ①² + ②² + ④² # Step 1

$$x^2+y^2+(z-L_1)^2 = R^2(c_1^2c_2^2 + s_1^2c_2^2 + s_2^2)$$
$$= R^2[c_2^2(c_1^2+s_1^2)+s_2^2]$$
$$= R^2[c_2^2+s_2^2]$$
$$x^2+y^2+(z-L_1)^2 = R^2$$
$$R = \gamma\sqrt{x^2+y^2+(z-L_1)^2}, \quad \gamma \in \{-1,1\} \quad -⑤$$
$$L_2+L_3+q_3 = \gamma\sqrt{x^2+y^2+(z-L_1)^2}$$
$$q_3 = \gamma\sqrt{x^2+y^2+(z-L_1)^2} - L_2 - L_3$$

จากสมการ ① ② # Step 2

จะได้

① ² + ② ²

$$x^2+y^2 = R^2[c_2^2(c_1^2+s_1^2)]$$
$$x^2+y^2 = R^2c_2^2$$
$$c_2 = \frac{\gamma\sqrt{x^2+y^2}}{R}, \quad \gamma \in \{-1,1\}$$
$$s^2+c^2 = 1 \Rightarrow s = \pm\sqrt{1-c^2}$$
$$s_2 = \gamma\sqrt{1-c^2}$$
$$q_2 = \operatorname{atan2}(s_2, c_2)$$

จากสมการ ① ②

$$X = Rc_1c_2$$
$$Y = Rs_1c_2$$
$$c_1 = \frac{Y}{Rc_2}$$
$$s_1 = \frac{X}{Rc_2}$$

Step 3

$$q_1 = \operatorname{atan2}(s_1, c_1)$$

Variable Constant.

$L_1 = 0.50$

$L_2 = 0.10$

$L_3 = 0.40$

q_3 limit = 0.5